

Quantifying Uncertainty in Marathon Finish Time Predictions

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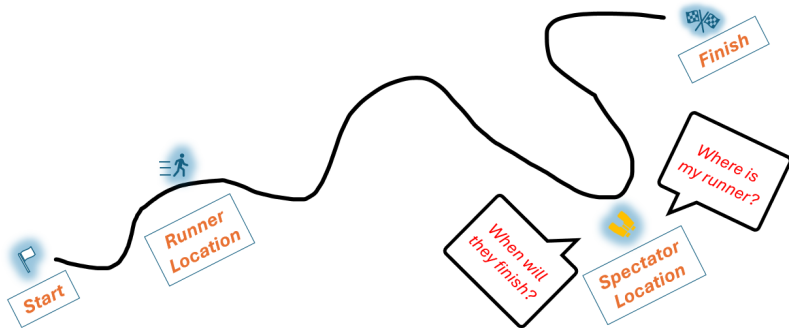


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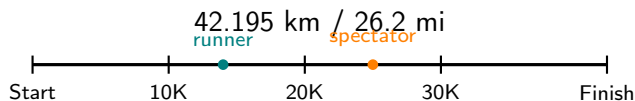
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- ▶ Modeling Marathon Finish Times
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A marathon prediction problem

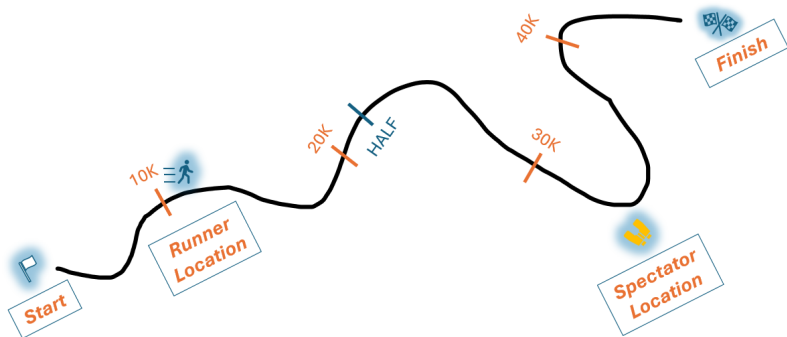


A marathon prediction problem



- ▶ Marathons are long-distance races: 42.195 km / 26.2 miles
- ▶ Large races include tens of thousands of runners moving at different paces
- ▶ Spectators often see a runner once and rely on intermediate splits to estimate finish time
- ▶ **Question:** given partial split information, when will this runner finish?

Intermediate splits and real-time prediction



Intermediate splits and real-time prediction

- ▶ Bib/chip timing records each runner's progress
- ▶ Common split points: 5K, 10K, 15K, 20K, half, 25K, 30K, 35K, 40K
- ▶ These splits are often available live through race websites or apps



Traditional estimate:

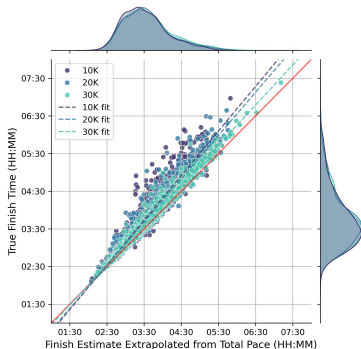
Extrapolation of current average pace

Stage	Time	Calculation	Prediction
After 10K	00:40:00	$\times 42.2/10.0$	02:48:48
After 20K	01:21:45	$\times 42.2/20.0$	02:52:30
After 30K	02:06:45	$\times 42.2/30.0$	02:58:18
Finish	03:01:20	—	03:01:20

- ▶ Baseline assumption: average pace so far will hold for the rest of the race
- ▶ Simple and interpretable, but ignores how races usually unfold

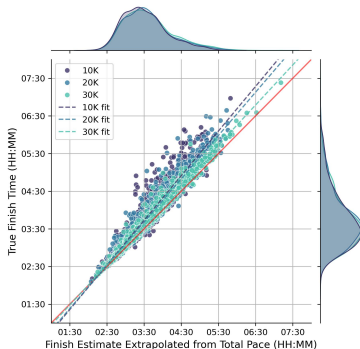
Traditional estimate = Extrapolation

- **Problem 1:** runners often slow in the second half; age, sex, course, and recent pace can all matter



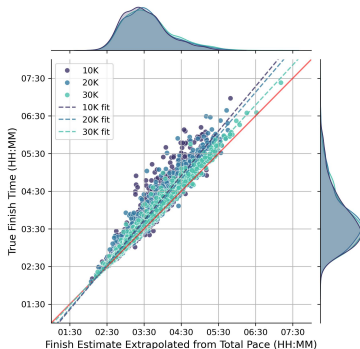
Traditional estimate = Extrapolation

- ▶ **Problem 1:** runners often slow in the second half; age, sex, course, and recent pace can all matter
- ▶ **Problem 2:** a single point estimate does not tell us how confident we should be



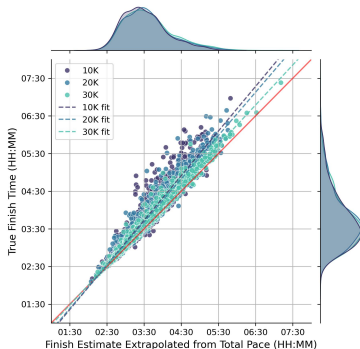
Traditional estimate = Extrapolation

- ▶ **Problem 1:** runners often slow in the second half; age, sex, course, and recent pace can all matter
- ▶ **Problem 2:** a single point estimate does not tell us how confident we should be
 - ▶ We should be less certain at 5K than 35K



Traditional estimate = Extrapolation

- ▶ **Problem 1:** runners often slow in the second half; age, sex, course, and recent pace can all matter
- ▶ **Problem 2:** a single point estimate does not tell us how confident we should be
 - ▶ We should be less certain at 5K than 35K
- ▶ **Goal:** improve point predictions *and* quantify finish-time uncertainty as the race unfolds.



Main Contribution

Current paper's focus

We implement a hierarchical Bayesian linear regression model that incorporates split-level race context and returns a posterior predictive distribution for each runner's finish time.

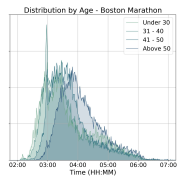
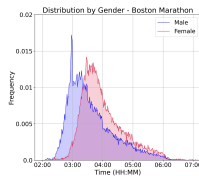
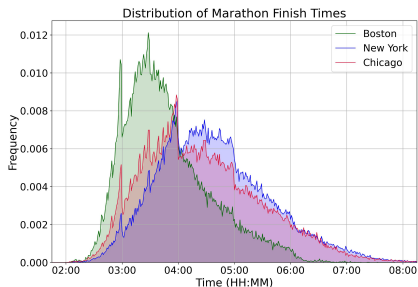
Data Collection

- ▶ Data come from the three U.S. World Marathon Majors
 - ▶ Boston, New York, Chicago
- ▶ Years: 2021–2024
- ▶ Data consist of all finishers:
 - ▶ Age
 - ▶ Gender
 - ▶ In-race splits
 - ▶ Finish time



Dataset Structure

- ▶ Each runner contributes split observations for the race
- ▶ Finish time is transformed into finish pace (m/s) for modeling, then converted back to time
- ▶ Primary pace features:
 - ▶ `total_pace`: pace from start to current split
 - ▶ `curr_pace`: pace over the most recent 5K segment
- ▶ Context features: gender, age, and course indicators



Training and test design

- ▶ Training set: random stratified sample of 30000 observations from 2021–2023
- ▶ Test set: randomly selected 30000 observations from 2024
 - ▶ Subsampling keeps Bayesian computation manageable
 - ▶ Kolmogorov–Smirnov checks suggest the subsamples are representative of the larger data distributions

Baseline and candidate models

Model	Predictors
BL	Traditional extrapolation from current average pace
S1	total pace + race
S2	total pace + gender + age + race
S3	total pace + current pace + race
S4	total pace + current pace + gender + age + race

- ▶ All Bayesian models outperform baseline
- ▶ S4 is the final working model emphasized in the paper
- ▶ Adding current pace captures recent acceleration or slowdown

Hierarchical Bayesian Regression (S1 and S2)

- Basic regression target: finish pace for runner i at split s

$$y_{is} \sim \mathcal{N}(X_i \beta_s, \sigma_s)$$

$$\beta_s \sim \mathcal{N}_D(0, I_D)$$

$$\sigma_s \sim \text{Half-Cauchy}(0, 1)$$

- Split-specific parameters allow predictions to change by distance into the race
- Hierarchical structure shares information across splits instead of fitting each split in isolation

Introducing Current Pace (S3 and S4)

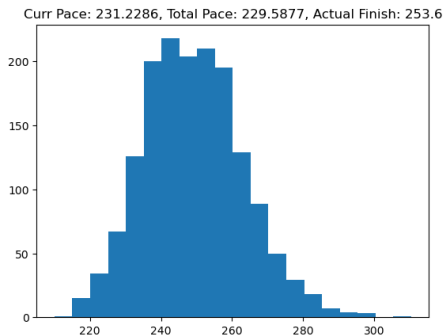
- ▶ At 5K, total_pace and curr_pace are identical
- ▶ To avoid convergence problems, current pace is separated for S3/S4 after 5K:

$$y_{is} \sim \mathcal{N}(X_i\beta_s + x_{ic}\gamma_s, \sigma_s)$$

- ▶ x_{ic} is current (previous split) 5K pace for runner i
- ▶ γ_s captures how recent pace matters at each race stage
 - ▶ γ_s also has a standard normal prior

Computation

- ▶ Implemented in `rstan` using Stan's Bayesian modeling framework
- ▶ Sampling uses NUTS/Hamiltonian Monte Carlo
- ▶ Four chains; 800 samples per chain; 400 warmup
- ▶ Model sizes: $S1 = 40$ parameters, $S2 = 56$, $S3 = 47$, $S4 = 63$

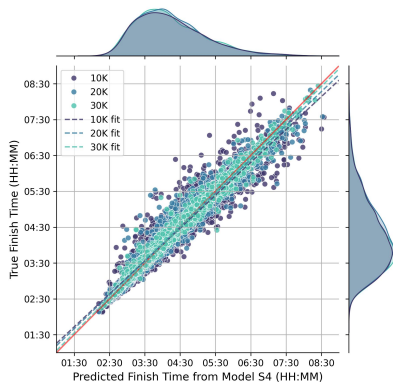
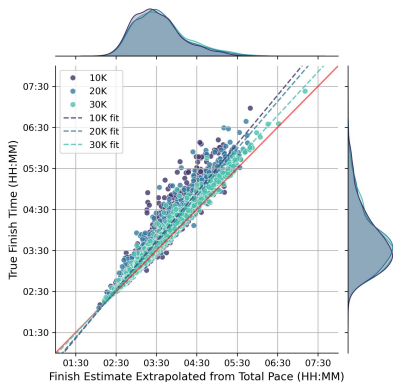


Solving Problem 1: Improve Prediction

	MAE				
Distance	BL	S1	S2	S3	S4
5K	22.027	17.085	16.935	17.089	16.937
10K	20.374	15.322	15.111	14.394	14.193
15K	19.2	13.745	13.503	12.179	11.948
20K	17.188	11.403	11.103	9.423	9.228
25K	13.287	9.057	8.798	6.741	6.614
30K	9.492	6.694	6.518	4.567	4.505
35K	5.175	3.927	3.76	2.547	2.519
40K	1.162	1.071	1.036	0.761	0.736
Overall MAE	13.517	9.808	9.616	8.484	8.356
Overall R^2	0.868	0.929	0.931	0.936	0.937

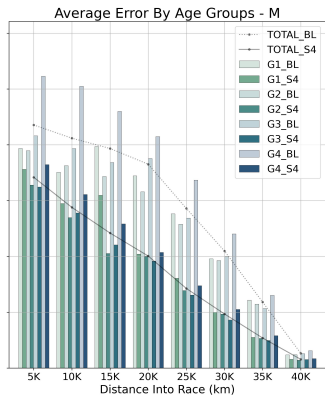
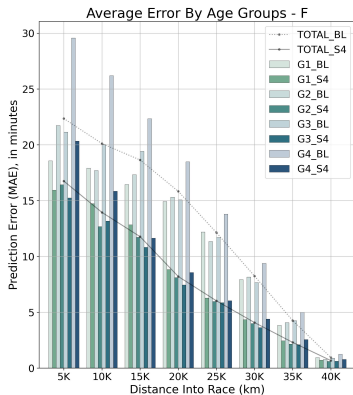
Solving Problem 1: Improve Prediction

- Predictions are now balanced throughout the race instead of overly optimistic



Performance Across Demographics

- S4 model performs best (and significantly outperforms baseline) across age and gender

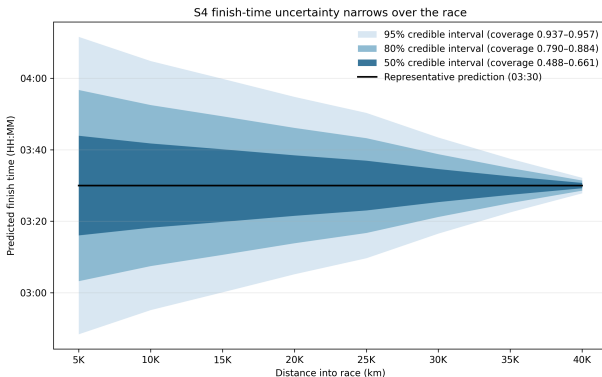


Inference

	α : Intercept		β_1 : total_pace		β_2 : male		β_3 : age_t	
Distance	Mean	Std	Mean	Std	Mean	Std	Mean	Std
5K	-0.153	0.0292	1.0067	0.007	-0.0433	0.008	-0.0768	0.0313
10K	-0.1232	0.0258	-0.0065	0.0489	-0.0502	0.0064	-0.0542	0.0272
15K	-0.1389	0.0209	-0.0042	0.0358	-0.0588	0.0056	0.0171	0.0238
20K	-0.0118	0.0183	0.2127	0.0219	-0.0561	0.0047	0.0473	0.0197
25K	0.0115	0.0138	0.4704	0.0133	-0.0406	0.0036	0.0247	0.0155
30K	0.0689	0.0094	0.6423	0.0075	-0.0207	0.0026	-0.0005	0.0105
35K	0.0726	0.0052	0.7885	0.0037	-0.0093	0.0014	-0.0124	0.0058
40K	0.0183	0.0015	0.948	0.001	-0.0013	0.0004	-0.0155	0.0017
	β_4 : race_nyc		β_5 : race_chi		γ : curr_pace		σ : sigma	
Distance	Mean	Std	Mean	Std	Mean	Std	Mean	Std
5K	-0.0937	0.0097	-0.0044	0.0092			0.2157	0.0025
10K	-0.1001	0.008	0.0266	0.0079	1.0025	0.0475	0.1782	0.0021
15K	-0.0536	0.0066	-0.0233	0.0066	1.0056	0.035	0.1543	0.0018
20K	-0.0326	0.0059	-0.0293	0.0055	0.7621	0.0207	0.129	0.0015
25K	0.0349	0.0049	-0.0036	0.0044	0.5007	0.0122	0.1066	0.0011
30K	-0.0488	0.0032	-0.027	0.0031	0.3326	0.0067	0.0706	0.0008
35K	-0.0216	0.0018	-0.0304	0.0017	0.1943	0.0033	0.0392	0.0004
40K	0.0021	0.0005	0.0037	0.0005	0.0485	0.0009	0.0115	0.0001

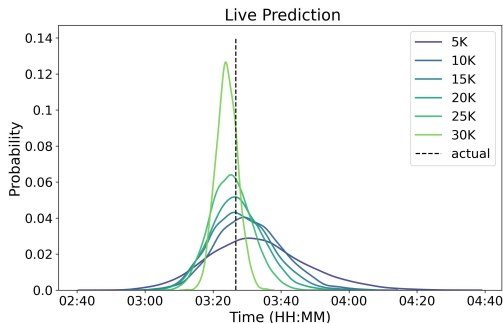
Solving Problem 2: Uncertainty Assessment

- Benefits of Being Bayesian: Posterior predictive distributions provide credible intervals
 - 50%, 80%, and 95% intervals are wide early and narrow near the finish, while coverage matches expectation



Solving Problem 2: Uncertainty Assessment

- Individual runners can visualize how the predictive distribution of their finish times evolves over each split



- This can help runners/trainers assess if they are speeding up/slowing down/maintaining pace

Summary and Future Work

- ▶ Key takeaways:
 - ▶ Traditional extrapolation is overconfident and context-free
 - ▶ Hierarchical Bayesian regression improves finish-time prediction across race splits
 - ▶ Performance in-line with previous work by Pradier et al. (2016) on the New York City marathons, but provides more balanced bias (Pradier et al. reported strictly positive bias—similar to issue with extrapolation)
 - ▶ Final model incorporates total pace, current pace, demographics, race context, and produces credible intervals that naturally narrow as runner approaches finish

Summary and Future Work

- ▶ Next Steps:
 - ▶ Explore additional covariate effects to improve model performance (e.g. runner experience or weather)
 - ▶ Explore mixed-effects framework (allowing individual runner random effects)
 - ▶ Using more years of data and estimating individual runner aging effects could also improve performance
 - ▶ Better study effects of subsampling and complete-case filtering (extending to dropped-out runners or runners with missing splits)
 - ▶ Continue improving R Shiny app for interactive prediction

Application for Real-Time Prediction (WIP)

- ▶ User selects a marathon
- ▶ User enters splits sequentially
- ▶ App updates predicted finish-time distribution after each split
- ▶ Output: median prediction, credible intervals, and graphic visualizing posterior predictive distribution



QUESTIONS?

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